KAIZEN MATHS

# Lesson starter

**Volumes of Revolution**

Expanding Before Integrating

Name _____

**S1**

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**S2**

Expand the squared bracket before integrating.

The region under $y = 3(x + 1)$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

**S3**

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**S4**

Expand the squared bracket before integrating.

The region under $y = 2(x - 2)$ from $x = 2$ to $x = 5$ is rotated about the x-axis. Find the volume.



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ALG-3EBB-L2 · S

Show your working

KAIZEN MATHS

# Exit ticket

**Volumes of Revolution**

Expanding Before Integrating

Name

**E1**

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**E2**

Expand the squared bracket before integrating.

The region under $y = 2(x - 1)$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

**E3**

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.



KAIZEN MATHS

Exit ticket

Volumes of Revolution

Expanding Before Integrating

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The region under $y = x^2 + 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

E2

Expand the squared bracket before integrating.

The region under $y = 2(x - 1)$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L2 · E

Show your working

KAIZEN MATHS

Exit ticket

Volumes of Revolution

Expanding Before Integrating

Name

E1

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

E2

Expand the squared bracket before integrating.

The region under $y = 2(x - 1)$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L2 · E

Show your working

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Exit ticket

Volumes of Revolution

Expanding Before Integrating

Name

E1

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

E2

Expand the squared bracket before integrating.

The region under $y = 2(x - 1)$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

E3

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

ALG-3EBB-L2 · E

Show your working

# Teacher answers & working

**Volumes of Revolution · Expanding Before Integrating**

ALG-3EBB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S1

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{202\pi}{5}$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

Substitute $y = x^2 + 3$: $V = \pi \int_0^2 (x^2 + 3)^2 \, dx$.

Expand before integrating: $(x^2 + 3)^2 = x^4 + 6x^2 + 9$.

So $V = \pi \int_0^2 (x^4 + 6x^2 + 9) \, dx$.

Integrate term by term: $\int (x^4 + 6x^2 + 9) \, dx = \dfrac{1}{5}x^5 + 2x^3 + 9x$.

Evaluate: $\left[\dfrac{1}{5}x^5 + 2x^3 + 9x\right]_0^2 = \dfrac{202}{5}$.

Therefore $V = \dfrac{202\pi}{5}$.

## S2

Expand the squared bracket before integrating.

The region under $y = 3(x + 1)$ from $x = 0$ to $x = 3$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 189\pi$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

The radius is $y = 3(x + 1)$, so $V = \pi \int_0^3 [3(x + 1)]^2 \, dx$.

Rewrite the radius as $3x + 3$.

Square the bracket: $(3x + 3)^2 = 9x^2 + 18x + 9$.

Integrate: $\int (9x^2 + 18x + 9) \, dx = 3x^3 + 9x^2 + 9x$.

Evaluate the definite integral: $\left[3x^3 + 9x^2 + 9x\right]_0^3 = 189$.

Therefore $V = 189\pi$.

# Teacher answers & working

**Volumes of Revolution · Expanding Before Integrating**

ALG-3EBB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## S3

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 2$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{376\pi}{15}$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

Substitute $y = x^2 + 2$: $V = \pi \int_0^2 (x^2 + 2)^2 \, dx$.

Expand before integrating: $(x^2 + 2)^2 = x^4 + 4x^2 + 4$.

So $V = \pi \int_0^2 (x^4 + 4x^2 + 4) \, dx$.

Integrate term by term: $\int (x^4 + 4x^2 + 4) \, dx = \dfrac{1}{5}x^5 + \dfrac{4}{3}x^3 + 4x$.

Evaluate: $\left[\dfrac{1}{5}x^5 + \dfrac{4}{3}x^3 + 4x\right]_0^2 = \dfrac{376}{15}$.

Therefore $V = \dfrac{376\pi}{15}$.

## S4

Expand the squared bracket before integrating.

The region under $y = 2(x - 2)$ from $x = 2$ to $x = 5$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = 36\pi$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

The radius is $y = 2(x - 2)$, so $V = \pi \int_2^5 [2(x - 2)]^2 \, dx$.

Rewrite the radius as $2x - 4$.

Square the bracket: $(2x - 4)^2 = 4x^2 - 16x + 16$.

Integrate: $\int (4x^2 - 16x + 16) \, dx = \dfrac{4}{3}x^3 - 8x^2 + 16x$.

Evaluate the definite integral: $\left[\dfrac{4}{3}x^3 - 8x^2 + 16x\right]_2^5 = 36$.

Therefore $V = 36\pi$.

# Teacher answers & working

**Volumes of Revolution · Expanding Before Integrating**

ALG-3EBB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## E1

Expand the squared radius before integrating.

The region under $y = x^2 + 2$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{83\pi}{15}$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

Substitute $y = x^2 + 2$: $V = \pi \int_0^1 (x^2 + 2)^2 \, dx$.

Expand before integrating: $(x^2 + 2)^2 = x^4 + 4x^2 + 4$.

So $V = \pi \int_0^1 (x^4 + 4x^2 + 4) \, dx$.

Integrate term by term: $\int (x^4 + 4x^2 + 4) \, dx = \dfrac{1}{5}x^5 + \dfrac{4}{3}x^3 + 4x$.

Evaluate: $\left[\dfrac{1}{5}x^5 + \dfrac{4}{3}x^3 + 4x\right]_0^1 = \dfrac{83}{15}$.

Therefore $V = \dfrac{83\pi}{15}$.

## E2

Expand the squared bracket before integrating.

The region under $y = 2(x - 1)$ from $x = 1$ to $x = 3$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{32\pi}{3}$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

The radius is $y = 2(x - 1)$, so $V = \pi \int_1^3 [2(x - 1)]^2 \, dx$.

Rewrite the radius as $2x - 2$.

Square the bracket: $(2x - 2)^2 = 4x^2 - 8x + 4$.

Integrate: $\int (4x^2 - 8x + 4) \, dx = \dfrac{4}{3}x^3 - 4x^2 + 4x$.

Evaluate the definite integral: $\left[\dfrac{4}{3}x^3 - 4x^2 + 4x\right]_1^3 = \dfrac{32}{3}$.

Therefore $V = \dfrac{32\pi}{3}$.

# Teacher answers & working

**Volumes of Revolution · Expanding Before Integrating**

ALG-3EBB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

## E3

Expand the squared radius before integrating.

The region under $y = x^2 + 3$ from $x = 0$ to $x = 1$ is rotated about the x-axis. Find the volume.

**Answer:**

$V = \dfrac{56\pi}{5}$

**Working:**

Use $V = \pi \int_a^b y^2 \, dx$.

Substitute $y = x^2 + 3$: $V = \pi \int_0^1 (x^2 + 3)^2 \, dx$.

Expand before integrating: $(x^2 + 3)^2 = x^4 + 6x^2 + 9$.

So $V = \pi \int_0^1 (x^4 + 6x^2 + 9) \, dx$.

Integrate term by term: $\int (x^4 + 6x^2 + 9) \, dx = \dfrac{1}{5}x^5 + 2x^3 + 9x$.

Evaluate: $\left[\dfrac{1}{5}x^5 + 2x^3 + 9x\right]_0^1 = \dfrac{56}{5}$.

Therefore $V = \dfrac{56\pi}{5}$.